

University of Information Technology & Sciences (UITS)

Department of Computer Science & Engineering
BSc. in CSE — 4th Year, 1st Semester | Spring 2026
CSE 413: Simulation & Modeling Lab
Section: 7A, 7B

Skill Mapping and Justification Form

CO–PO Mapping, Complex Engineering Problem (CEP) and Knowledge Profile (KP)
Justification Report

Assignment/Project Title: “Designing a Simulation-Based Smart Traffic Analysis System for Urban Road Management”

No.	COs	POs	BT	CP/WP	CA/EA	KP/WK
CO1	Define various modelling terms by analyzing the given system or data.	PO(a)	L1, L2, L4	–	–	K2
CO2	Classify various simulation models and their application in simulation.	PO(b)	L2, L3, L4	–	–	K2, K3
CO3	Construct a model for a given set of data and examine its validity.	PO(c)	L3, L4, L5, L6	P1, P2, P3	–	K2, K3, K5
CO4	Analyze output data produced by a model and test the validity of the model.	PO(d)	L4, L5, L6	P1, P2, P3	–	K8

*COs- Course Outcome; POs -Program Outcome; CP/WP- Complex Problems; CA/EA- Complex Activities; KP/WK- Knowledge Profile.

Justification of Program Outcome (PO) through Course Outcome (COs)

COs No.	POs	Justification
CO1	PO1 (Engineering Knowledge)	The lab activities require students to apply fundamental concepts of mathematics, statistics, probability, and computer science to understand modeling terminology and system behavior. Students analyze system variables, entities, events, states, and performance measures while utilizing theoretical concepts

COs No.	POs	Justification
		necessary for representing real-world systems mathematically and computationally.
CO2	PO2 (Problem Analysis)	Students analyze different types of simulation models including deterministic, stochastic, discrete-event, continuous, and Monte Carlo models to investigate complex engineering problems. The lab requires examination of system characteristics, identification of suitable modeling approaches, and analytical comparison of alternative simulation techniques before selecting the most appropriate method for a given application.
CO3	PO3 (Design/Development of Solutions)	The lab requires students to design and construct simulation models for real-world systems based on given datasets and problem specifications. Students develop modeling structures, define system parameters, implement simulation logic, and validate model behavior to produce reliable and efficient solutions that satisfy functional requirements and support decision-making processes.
CO4	PO4 (Investigation)	Students investigate the behavior and performance of simulated systems through output analysis and statistical validation techniques. The lab involves conducting experiments, interpreting simulation results, testing model accuracy, and evaluating the validity and reliability of models using analytical and research-oriented approaches to support engineering conclusions.

Justification of Complex Engineering Problem (P) and Knowledge Profile (KP)

Attributes	Justification
P1: Depth of Knowledge	Solving the smart traffic analysis problem requires in-depth engineering knowledge and a first-principles approach. Students must apply theory-based engineering fundamentals (K3), including probability distributions, random number generation, statistical analysis, Monte Carlo simulation, and hypothesis testing to model uncertain traffic behavior. The assignment further requires engineering design knowledge (K5) to develop a reliable computational traffic simulation model capable of generating realistic traffic scenarios and supporting traffic management decisions. Model validation and investigation activities also involve research-oriented statistical techniques and analytical methods, thereby engaging students with research knowledge and methodologies (K8).

Attributes	Justification
P2: Range of Conflicting Requirements	Students must balance conflicting requirements such as simulation accuracy versus computational efficiency, reliability versus model complexity, realistic representation of traffic uncertainty versus simplicity of assumptions, and scalability versus execution cost. Since multiple valid approaches are possible, including random number-based simulation, probability distribution modeling, Monte Carlo simulation, and statistical validation methods, students are required to evaluate alternatives, justify their selected approach, and make engineering trade-off decisions to obtain an effective solution.
P3: Depth of Analysis Required	The assignment has no straightforward solution and requires originality and abstract thinking in formulating suitable models. Students must analyze traffic data, identify appropriate probability distributions, generate simulated traffic scenarios, compare simulated and observed traffic behaviors, and validate the model statistically. They are further required to investigate assumptions, limitations, sources of error, and possible improvements. Therefore, the problem demands detailed analytical reasoning and engineering judgment rather than the application of memorized procedures.

Justification of Knowledge Profile (KP) [because KP2 Cannot be resolved by P1]

Knowledge Profile	Justification
K2 – Conceptual Understanding of Engineering Sciences	The assignment requires conceptual understanding of probability theory, statistical measures, random variables, probability distributions, random sampling, Monte Carlo simulation, and hypothesis testing to represent uncertain traffic systems computationally. Students conceptually analyze vehicle arrival patterns, waiting-time behavior, and traffic-flow characteristics while understanding how different probability models and statistical techniques capture real-world randomness and support simulation-based decision making.